

ATUL KUMAR

Software Engineer | Bangalore, India

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SUMMARY

Software Engineer with **3+ years of professional experience** building production systems in financial services. Specialized in backend development, distributed systems, microservices architecture, cloud infrastructure (AWS), data pipelines (Apache Spark, Kafka), and machine learning (LangGraph, SageMaker, Retrieval-Augmented Generation). Strong track record of cross-functional collaboration, stakeholder management, and end-to-end ownership across system design, software development, and AI/ML integration.

CORE COMPETENCIES

Leadership & Collaboration: Cross-Functional Collaboration, Stakeholder Management, Project Ownership, Mentoring, Technical Communication, Team Coordination

Problem Solving: Analytical Thinking, Root-Cause Analysis, Critical Thinking, Decision-Making, Strategic Planning, Continuous Improvement

TECHNICAL SKILLS

Programming Languages: Java, Python, C++, SQL, JavaScript, TypeScript

Backend & Software Development: Spring Boot, Flask, REST APIs, Microservices Architecture, Distributed Systems, System Design, Object-Oriented Programming (OOP), SOLID Principles, Server-Side Streaming, API Design

Data Engineering: Apache Spark 3, Apache Kafka, AWS Glue, Snowflake, Sybase IQ, ETL, Data Modeling, Data Warehousing, Near-Real-Time (NRT) Pipelines, Stream Processing

AI / Machine Learning: LangGraph, AWS SageMaker, Retrieval-Augmented Generation (RAG), Embeddings, Vector Search, Optical Character Recognition (OCR), Large Language Models (LLMs), Natural Language Processing (NLP), Machine Learning

Cloud & DevOps: AWS (Lambda, ECS, Step Functions, S3, Aurora, API Gateway, VPC, ELB, Route 53), Terraform, CloudFormation, Docker, CI/CD, Git, Linux, Agile, Scrum

Software Engineering Practices: Unit Testing, Integration Testing, Test-Driven Development (TDD), Code Review, Performance Optimization, Software Architecture, ReactJS, MongoDB, Elasticsearch

PROFESSIONAL EXPERIENCE

Goldman Sachs | Engineering Associate | Bangalore, India | Jan 2025 - Present

TINA - Travel Insights & Navigation Agent (AI)

- Deflected **80%** of travel & expense support queries, saving **1,700+ man-hours/year**, by building an internal AI agent that resolves policy questions end-to-end.
- Achieved average response latency under **20 seconds** across 40 policy documents via a SageMaker-backed Retrieval-Augmented Generation (RAG) pipeline orchestrated by a LangGraph multi-step retrieval agent.

Travel & Expense Platform

- Enabled streaming export of **5M+ live records per request** at ~50K records/sec within a 10-minute window via a server-side API, eliminating timeouts and OOM failures of the prior batch flow.
- Eliminated daily manual reconciliation across **80 global markets** by building an auto/manual voucher flow for Amex vendor payments with auto-generated ledgers, statements, and reconciliation.
- Halved the dedicated operations team from **2 people to 1** through automation of the reconciliation workflow.
- Met Goldman Sachs' **Tech Raise Bar** for the greenfield platform by authoring an end-to-end integration test suite spanning Step Functions, ECS, S3, and Aurora.

AI-Driven Invoice Lifecycle Management (Proof of Concept)

- Eliminated manual invoice data entry via an Optical Character Recognition (OCR) layer that pre-fills **100%** of form fields for human verification.
- Prevented duplicate payments across **\$2M+ daily invoice volume** by implementing a duplicate-detection layer over historical submissions.
- Built a machine-learning-driven reviewer-assignment engine routing invoices by reviewer expertise, calendar availability, and criticality, with cycle-time metrics feeding continuous improvement.

Country Exposure Report (FFIEC 009)

- Met tier-1 FFIEC 009 requirements across **12 upstream data sources** via a high-performance Spark 3 solution on AWS Glue, and cut regulatory-report build time by **~75%** through shared Spark libraries.

Slate - Inter-Affiliate Outsourcing Agreements

- Drove data-backed feature prioritization by integrating GS Analytics into the Slate frontend, capturing feature-level telemetry from **800 users** across 10 workflows in the agreement-creation pipeline.

Goldman Sachs | Engineering Analyst | Bangalore, India | Jun 2022 - Dec 2024

TIC B Report Uplift

- Cut data processing time by **83%** (from 12 hours to 2 hours) by re-architecting the pipeline on Apache Spark 3.
- Accelerated monthly data availability by **9 days** by removing legacy dependencies and rebuilding the data model, giving users 10+ extra business days before the regulatory deadline.
- Enabled exact reproduction of any historical calculation across **60 monthly reports** via a versioning system that retrieves precise inputs and logic for any past run.
- Fully automated the monthly close cycle, eliminating manual hand-offs across the end-to-end data pipeline.

TIC SLT / SHCA / SHLA Reports Uplift

- Advanced monthly data availability by **17 days** (18th to 1st of month) by shifting end-of-cycle batch processing to daily incremental processing.
- Cut data-quality issue resolution from **4-5 days** of manual email follow-up to near-real-time by building a UI over Kafka-driven near-real-time (NRT) pipelines that surfaces issues as they occur.

American Express | Technology Intern | Bangalore, India | Jan 2021 - Jun 2021

- Built an end-to-end natural-language query parser for Amex's ML Studio Logstash, translating plain-English questions into executable Logstash queries and removing the tool's learning curve for new users.

EDUCATION

Bachelor of Engineering (B.E.), Computer Science and Engineering | Punjab Engineering College, Chandigarh | Aug 2018 - Jun 2022 | CGPA 8.45 / 10

Class 12, CBSE | Springdale Public School, Kurukshetra | 2018 | 89.2%

Class 10, CBSE | Springdale Public School, Kurukshetra | 2016 | CGPA 10 / 10

ACHIEVEMENTS & LEADERSHIP

- Owned multiple regulatory reports end-to-end (incl. TIC-B) - raw data pipeline through published report.
- Organized PEC-ACM technical events at **PECfest** with 45+ participants across 15 teams.